

Technical Documentation & User Guide

Simulation & Analysis Suite for 1D Laminar Deflagrations in Type Ia Supernovae

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1. Project Overview & Architecture

This software suite automates the end-to-end parametric study workflow for 1D laminar deflagration fronts in stellar media (Type Ia Supernovae): from physical configuration generation to rigorous numerical analysis of steady flame speed and results visualization.

Repository Structure

```
laminar-deflagration/  
├── generate_simulations.py    # Configuration generator & SLURM job launcher  
├── compare_flame_positions.py # Flame position calculator & fitting/extrapolation  
├── robust_extrapolation.py    # Plateau detection & Aitken extrapolation / regression  
├── analyze_results.py         # Main CLI for analysis, benchmarks & plotting  
├── heatmap_and_3D_visualisation.py # Interactive/headless 2D/3D rendering tool (CSV)  
├── compare_csv.py             # Comparison utility between two CSV result files  
├── template_simulation/       # Fortran templates (app.F90, Makefile, run.genoa, plot.sh)  
├── parameter_study/          # Generated directory containing simulation folders  
└── parameter_study_results.csv # Auto-generated data cache
```

2. Physical Modeling & Self-Similar Scaling

To guarantee numerical regularity across the entire phase space without introducing theoretical bias from empirical lookup tables, the generation script `generate_simulations.py` computes self-similar spatial scales based on classical laminar flame theory:

$$\delta_{est}(\rho, X_{C12}) = \delta_{ref} \cdot (\rho / 10^9 \text{ g cm}^{-3})^{-0.75} \cdot (X_{C12} / 0.5)^{-1.0}$$

Where $\delta_{ref} = 2.2 \times 10^{-4} \text{ cm}$ represents the reference thermal thickness at $\rho = 10^9 \text{ g/cm}^3$ and C/O ratio = 0.5.

Mesh Adaptation & Composition Rules

- **Self-adaptive dynamic mesh:** The spatial domain size is set to $x_{2u} = 300 \cdot \delta_{est}$ with a fixed grid resolution $N_{x2} = 2048$. This strictly enforces $\Delta x = x_{2u} / N_{x2} \approx \delta_{est} / 6.83$, yielding 13 to 14 numerical cells within the flame front across all densities and compositions.
- **Ignition zone (x_t) & smoothing:** The initial burned thickness is set to $L_{ignition} = 1.0 \cdot \delta_{est}$ (i.e., $x_t = L_{ignition} / x_{2u}$). The tanh smoothing parameter `deltax` is directly scaled to the spatial resolution Δx .
- **Electron fraction (Y_e) & Neutronicisation:** To lower Y_e below 0.5 without altering the C/O ratio, spectator isotope ^{40}Ne ($Z=10$, $A=40$, $Z/A=0.25$) is incorporated:
$$X_{Ne40} = 4 \cdot (0.5 - Y_e), \quad X_{C12} = R_{CO} \cdot (1 - X_{Ne40}), \quad X_{O16} = (1 - R_{CO}) \cdot (1 - X_{Ne40}).$$

3. Detailed Description of Python Scripts

3.1. generate_simulations.py (Generation & Execution)

Master script for setting up the parameter space (ρ from 10^7 to 10^{10} g/cm³, C/O from 0.2 to 1.0, Y_e from 0.490 to 0.500). It automatically patches `app.F90` and `Makefile` files and submits SLURM jobs via `sbatch`.

```
$ python3 generate_simulations.py generate --dry-run
$ python3 generate_simulations.py generate --rho 1e9 --co 0.5 --ye 0.500
$ python3 generate_simulations.py run --all --generate
```

3.2. compare_flame_positions.py & robust_extrapolation.py (Speed Measurement)

These modules compute the average flame front position $r(t)$ at each HDF5 snapshot (`rprofs_n*.h5`) by combining the temperature midpoint T_{target} and the nuclear energy generation rate peak $e_{dot_{nuc}}$ (with a 90% domain boundary cutoff).

The asymptotic speed v_{sim} is extracted via non-linear exponential relaxation fitting:

$$r(t) = r_0 + v_{\infty} \cdot t + A \cdot \tau \cdot (1 - e^{-t/\tau})$$

Robustness Guarantee (Cross-Checking & Independence)

In case of exponential fit collapse/degeneracy, `robust_extrapolation.py` applies an independent cross-check (OLS sliding window + Aitken δ^2 extrapolation + plateau detection). If both methods diverge by more than 5%, a warning flag is raised in the summary table.

3.3. analyze_results.py (Main CLI & Analysis)

Primary post-processing analysis tool. Features an **automatic CSV caching system** (`parameter_study_results.csv` with 16 columns), reducing re-execution load times to < 0.1s.

Mode	Filters / Options	Description & Usage
VALIDATION	<code>--ye 0.5</code>	Compares v_{sim} against theoretical models by Timmes & Woosley (1992) and computes relative % errors.
EXPLORATION	<code>--ye-curves</code> , <code>--heatmap</code> , <code>--surface</code>	Analyzes $v(Y_e)$ sensitivity, generates 2D heatmaps $v(\rho, Y_e)$ or 3D surfaces.
FIXED YE	<code>--heatmap-co</code> , <code>--surface-co</code>	Displays 2D/3D dependencies $v(\rho, X_C)$ at fixed Y_e .

```
$ python3 analyze_results.py --ye 0.5 --only-table
$ python3 analyze_results.py --force --co 0.5 --heatmap
$ python3 analyze_results.py --csv export_custom.csv --only-table
```

3.4. heatmap_and_3D_visualisation.py & compare_csv.py (Visualization & Diagnostics)

- heatmap_and_3D_visualisation.py** : Generates interactive or headless 2D/3D figures (SSH compatible without X server using backend `Agg`) directly from a CSV file.
`python3 heatmap_and_3D_visualisation.py --heatmap results.csv --x ye --y rho --z v_sim`
- compare_csv.py** : Compares two CSV exports to assess algorithm modification impacts (e.g., 90% boundary cutoff). Isolates relative variations exceeding 1%.
`python3 compare_csv.py old_results.csv new_results.csv`

4. CLI Parameter Reference Guide

CLI Argument	Type	Default Value	Description & Role
<code>--rho</code>	Float	None	Filter on unburned fuel density [g/cm³].
<code>--co</code>	Float	None	Filter on C/O mass ratio: $X_{C12} / (X_{C12} + X_{O16})$.
<code>--ye</code>	Float	None	Filter on electron fraction Y_e .
<code>--force</code>	Flag	False	Forces complete re-computation of HDF5 snapshots, bypassing CSV cache.
<code>--only-table</code>	Flag	False	Prints only the synthetic ASCII table without opening graphical windows.
<code>--csv [PATH]</code>	String	None	Exports filtered or full results to the specified CSV path.